INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA

SALTLAKE

(School of University of Engineering and Management, Kolkata)

Degree: BCA, Stream: BCA

Term – II ,Third Semester Examination, October - 2025

Subject Code: BCACC302

Subject Name: Operating Systems

Full Marks: 30 Duration: 1 Hour

Part - A
Answer any 5 out of 6 questions
Each question carries 2 marks (2 × 5)

1. Explain the concept of paging in memory management. **2**

2. What is memory fragmentation? **2**

3. What is MMU in memory management? **2**

4. Define Claim Edge. **2**

5. Define Available and Allocation in Banker's Algorithm. **2**

6. Define Mutual Exclusion. **2**

Part - B
Answer any 2 out of 3 questions
Each question carries 5 marks (5 × 2)

7. Differentiate contiguous and non-contiguous memory allocation. **5**

8. Demonstrate Many to Many Multithreading concept. **5**

9. Describe paging and segmentation in memory management. **5**

Part - C
Answer any 1 out of 2 questions
Each question carries 10 marks (10 × 1)

10. Create the Safe Sequence of the following Snapshot at time T_0: for 5 processes P_0 through P_4 having 3 resource types A (10 instances), B (5instances, and C (7 instances). **10**

	Allocation	*Max*	*Available*
	A B C	*A B C*	*A B C*
P_0	0 1 0	7 5 3	3 3 2
P_1	2 0 0	3 2 2	
P_2	3 0 2	9 0 2	
P_3	2 1 1	2 2 2	
P_4	0 0 2	4 3 3	

11. A system uses 3 page frames for storing process pages in main memory. Assume that all the page frames are initially empty. Calculate the total number of page faults for FIFO while processing the page reference string given below **10**

4 , 7, 6, 1, 7, 6, 1, 2, 7, 2.